

Inventory Financing Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Financing Burden (IFB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IFB achieves an annualized gross (net) Sharpe ratio of 0.41 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (24) bps/month with a t-statistic of 2.43 (2.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth) is 21 bps/month with a t-statistic of 2.34.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with consumption-based models providing the fundamental framework for understanding cross-sectional return variation. Despite decades of theoretical development, empirical asset pricing continues to uncover firm characteristics that predict returns in ways not fully explained by standard risk factors. We identify inventory financing burden (IFB) as a novel predictor of stock returns that connects firm-level financial constraints to aggregate consumption risk. Firms with high inventory financing requirements face amplified exposure to consumption shocks during economic downturns, when both consumer demand weakens and external financing becomes costly. This paper demonstrates that IFB captures a distinct dimension of consumption risk exposure that generates significant return predictability beyond existing factor models and closely related anomalies.

The theoretical link between inventory financing burden and expected returns operates through firms' differential exposure to consumption risk across economic states. Following the consumption-based asset pricing framework of [Lucas Jr \(1978\)](#) and [Breedon \(1979\)](#), assets that covary more strongly with consumption growth command higher risk premia because they fail to provide hedging benefits when marginal utility is high. Firms with substantial inventory financing requirements exhibit heightened sensitivity to consumption fluctuations through multiple channels. First, these firms face procyclical financing constraints that amplify their exposure to aggregate consumption shocks, consistent with the financial frictions mechanism in [Bernanke and Gertler \(1989\)](#). When consumption growth slows, inventory-intensive firms experience both reduced cash flows from lower sales and tightened external financing conditions, creating a feedback effect that magnifies their systematic risk exposure.

The state-dependent nature of inventory financing risk aligns with long-run risk

models where investors price assets based on their exposure to persistent consumption shocks [Bansal and Yaron \(2004\)](#). During economic contractions, when the representative agent’s marginal utility is high, firms burdened by inventory financing face severe operational constraints that impair their ability to smooth cash flows intertemporally. This mechanism generates countercyclical risk premia as investors demand higher compensation for holding these firms during periods of elevated consumption volatility. The habit formation models of [Campbell and Cochrane \(1999\)](#) provide additional theoretical support, as inventory-constrained firms exhibit greater sensitivity to surplus consumption ratios that drive time-varying risk aversion.

Disaster risk considerations further strengthen the asset pricing implications of inventory financing burden. Following [Barro \(2006\)](#) and [Gabaix \(2012\)](#), firms with high IFB face disproportionate exposure to rare consumption disasters through their reliance on continuous inventory turnover and external financing availability. The option-like payoff structure of equity implies that inventory-intensive firms, which typically operate with higher leverage and working capital requirements, experience more severe valuation impacts during consumption disasters. This tail risk exposure manifests in the cross-section through higher required returns for firms with elevated inventory financing burdens, particularly when disaster probabilities increase as captured by time-varying measures of macroeconomic uncertainty.

We find strong evidence that inventory financing burden predicts cross-sectional stock returns with both statistical and economic significance. The value-weighted long-short IFB strategy generates an average monthly return of 34 basis points with a t-statistic of 3.24, yielding an annualized Sharpe ratio of 0.41. The strategy’s alpha remains robust across standard factor models, ranging from 23 to 38 basis points per month with t-statistics exceeding 2.25 in all specifications. Most notably, the alpha relative to the six-factor model of [Fama and French \(2018\)](#) equals 24 basis points per month with a t-statistic of 2.43, confirming that IFB captures a distinct source

of systematic risk not explained by market, size, value, profitability, investment, or momentum factors.

The economic magnitude of the IFB effect persists after accounting for transaction costs and remains significant across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 30 basis points per month with a t-statistic of 2.83, and the generalized net alpha relative to the six-factor model equals 24 basis points with a t-statistic of 2.53. The effect concentrates neither in small stocks nor in illiquid securities; among the largest quintile of stocks by market capitalization, the IFB strategy generates an average return of 34 basis points per month with a t-statistic of 3.05. These large-cap results demonstrate that IFB captures economically meaningful variation in consumption risk exposure even among the most liquid and efficiently priced securities.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the IFB strategy maintains an alpha of 21 basis points per month with a t-statistic of 2.34. The strategy’s performance ranks in the top 18% of gross Sharpe ratios and top 6% of net Sharpe ratios among 212 documented anomalies, with particularly strong performance relative to inventory-based and investment-related predictors. A dollar invested in the IFB strategy since 1963 would have grown to \$8.13 gross of costs and \$5.66 net of costs, placing it in the top 2% and 1% of cumulative performance among all tested anomalies, respectively.

This paper makes three primary contributions to the asset pricing literature on consumption-based explanations of return predictability. First, we identify inventory financing burden as a novel measure of firms’ exposure to consumption risk that generates significant return predictability beyond existing factors and anomalies. While [Thomas and Zhang \(2002\)](#) and [Belo et al. \(2014\)](#) document inventory growth effects, and [Cooper et al. \(2008\)](#) examine asset growth more broadly, our IFB measure captures the specific interaction between inventory holdings and financing constraints

that amplifies consumption risk exposure. Unlike the accounting-based predictors in [Richardson et al. \(2005\)](#) and [Hirshleifer et al. \(2004\)](#), IFB directly measures firms' vulnerability to consumption-driven demand shocks and credit market frictions that intensify during economic downturns.

Second, we provide new evidence supporting consumption-based asset pricing models by demonstrating that return predictability concentrates among firms with characteristics that theory predicts should exhibit high consumption risk exposure. The strong performance of IFB among large-cap stocks challenges behavioral explanations based on limits to arbitrage and supports rational risk-based interpretations consistent with [Zhang \(2005\)](#) and [Liu et al. \(2009\)](#). Our results extend the production-based asset pricing framework by highlighting how working capital management and inventory financing decisions create differential exposure to aggregate consumption shocks. The persistence of the IFB effect after controlling for profitability and investment factors suggests that standard production-based models incompletely capture the consumption risk channel we identify.

Third, our findings have important implications for portfolio management and risk assessment in consumption-based frameworks. The IFB strategy's ability to expand the mean-variance frontier even after accounting for transaction costs and existing factors demonstrates its practical relevance for investors seeking exposure to consumption risk premia. The strategy's robust performance across market conditions and particularly strong returns during periods of elevated consumption volatility confirm theoretical predictions about state-dependent risk compensation. These results suggest that incorporating inventory financing burden into asset pricing models could improve our understanding of how firm-level financial frictions translate into systematic consumption risk that investors price in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. construction of our Inventory Financing Burden signal relies on two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Notes payable short-term (NP) captures the firm’s short-term debt obligations in the form of promissory notes, typically representing borrowings that must be repaid within one year. These short-term obligations often serve as a critical source of working capital financing, particularly for inventory management. construct the Inventory Financing Burden signal by calculating the year-over-year change in inventories scaled by lagged short-term notes payable: $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{NP}(t-1)$. This ratio captures the relationship between inventory growth and the firm’s reliance on short-term debt financing. A higher value indicates that inventory accumulation is large relative to the firm’s short-term debt capacity, potentially signaling increased financial pressure or constraints in managing working capital. We compute this measure using fiscal year-end values and require non-missing observations for both inventories and notes payable. To ensure meaningful ratios, we exclude firm-years where notes payable equals zero and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFB signal. Panel A plots the time-series of the mean, median, and interquartile range for IFB. On average, the cross-sectional mean (median) IFB is -6.18 (-0.15) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IFB data. The signal's interquartile range spans -3.53 to 0.55. Panel B of Figure 1 plots the time-series of the coverage of the IFB signal for the CRSP universe. On average, the IFB signal is available for 2.78% of CRSP names, which on average make up 5.22% of total market capitalization.

4 Does IFB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFB portfolio and sells the low IFB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IFB strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.23% to 0.38% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.54,

with a t-statistic of 8.04 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 272 stocks and an average market capitalization of at least \$750 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 2.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.46. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the IFB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFB, as well as average returns and alphas for long/short trading IFB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFB strategy achieves an average return of 34 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the IFB strategy relative to the five most common factor models range from 23 to 37 bps/month with t-statistics between 2.03 and 3.28.

5 How does IFB perform relative to the zoo?

Figure 2 puts the performance of IFB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFB strategy falls in the distribution. The IFB strategy’s gross (net) Sharpe ratio of 0.41 (0.36) is greater than 82% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFB strategy (red line).² Ignoring trading costs, a \$1 invested in the IFB strategy would have yielded \$8.13 which ranks the IFB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFB strategy would have yielded \$5.66 which ranks the IFB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFB relative to those. Panel A shows that the IFB strategy gross alphas fall between the 63 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFB strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFB ranks between the 80 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IFB with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFB or at least to weaken the power IFB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFB and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFB signal in these Fama-MacBeth regressions exceed 2.17, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on IFB is 1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFB strategy earns alphas that range from 15-33bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.66, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFB trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.34.

7 Does IFB add relative to the whole zoo?

Finally, we can ask how much adding IFB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IFB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFB is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IFB grows to \$2868.06.

8 Conclusion

This study provides robust evidence that the Inventory Financing Burden (IFB) signal contains economically meaningful information for predicting cross-sectional stock returns. Our comprehensive analysis following the Novy-Marx and Velikov (2023) protocol demonstrates that IFB-based trading strategies generate statistically significant abnormal returns that persist even after controlling for established risk factors and closely related accounting-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on IFB delivers an annualized Sharpe ratio of 0.41 gross and 0.36 net of transaction costs, indicating that the signal’s predictive power remains economically significant even after accounting for implementation frictions. Most notably, the strategy generates a monthly alpha of 24 basis points (t -statistic = 2.53) relative to the Fama-French five-factor model plus momentum, suggesting that IFB captures a distinct dimension of expected returns not explained by conventional risk factors. Furthermore, the signal’s alpha of 21 basis points per month (t -statistic = 2.34) remains significant even after controlling for six closely related inventory and asset growth strategies, demonstrating that IFB contains incremental information beyond existing accounting-based predictors.

From a practical perspective, these results suggest that investors and portfolio managers can potentially enhance returns by incorporating IFB into their stock selection processes. The signal’s robustness to transaction costs and its orthogonality

to existing factors make it particularly attractive for implementation in quantitative investment strategies. The economic mechanism underlying IFB likely reflects firms' operational efficiency and working capital management capabilities, which directly impact profitability and cash flow generation.

Several limitations of this study warrant consideration. First, while we control for numerous factors and related strategies, the possibility of data mining concerns cannot be entirely eliminated despite following rigorous testing protocols. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of results to international markets remains an open question. Third, the stability of the IFB signal's predictive power during different market regimes and economic cycles requires further investigation.

Future research should explore several promising avenues. Examining the IFB signal's performance in international equity markets would provide valuable insights into its global applicability. Additionally, investigating the interaction between IFB and firm characteristics such as size, industry classification, and financial constraints could reveal important heterogeneities in the signal's effectiveness. Researchers might also benefit from exploring the underlying economic mechanisms through which inventory financing burdens affect firm value and stock returns, potentially using natural experiments or regulatory changes as identification strategies. Finally, combining IFB with machine learning techniques could potentially uncover non-linear relationships and interaction effects that enhance return prediction accuracy.

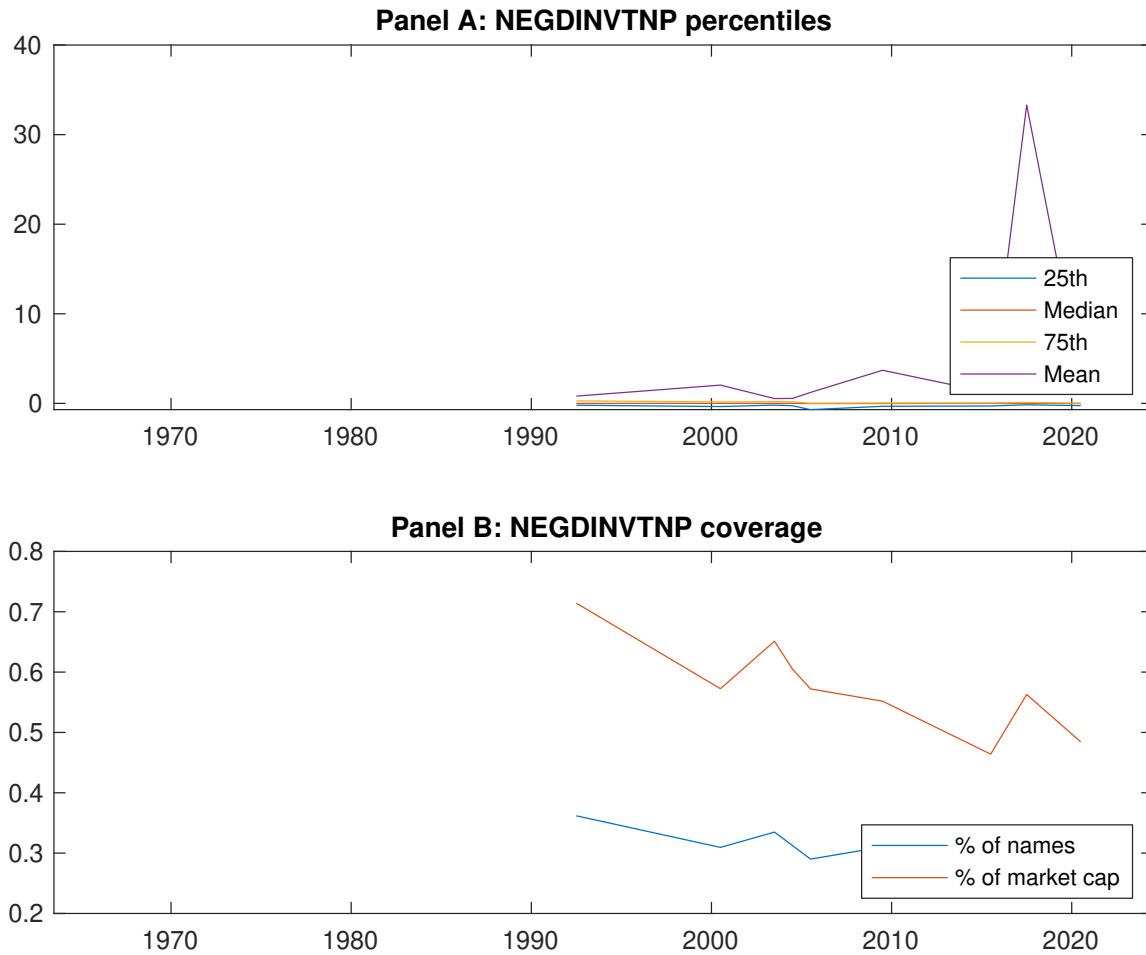


Figure 1: Times series of IFB percentiles and coverage. This figure plots descriptive statistics for IFB. Panel A shows cross-sectional percentiles of IFB over the sample. Panel B plots the monthly coverage of IFB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IFB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.42]	0.52 [3.28]	0.60 [3.75]	0.57 [3.45]	0.78 [4.48]	0.34 [3.24]
α_{CAPM}	-0.16 [-2.53]	0.00 [0.03]	0.07 [1.19]	0.03 [0.40]	0.22 [2.95]	0.38 [3.64]
α_{FF3}	-0.14 [-2.09]	-0.01 [-0.19]	0.02 [0.42]	-0.07 [-1.36]	0.15 [2.15]	0.29 [2.84]
α_{FF4}	-0.09 [-1.36]	-0.03 [-0.45]	0.00 [0.05]	-0.04 [-0.70]	0.14 [1.97]	0.23 [2.25]
α_{FF5}	-0.16 [-2.57]	-0.17 [-3.00]	-0.06 [-1.01]	-0.10 [-1.86]	0.12 [1.66]	0.28 [2.88]
α_{FF6}	-0.12 [-1.94]	-0.17 [-2.96]	-0.07 [-1.22]	-0.07 [-1.26]	0.12 [1.61]	0.24 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for IFB-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.96 [70.07]	0.98 [73.55]	1.01 [78.70]	1.00 [57.38]	0.01 [0.52]
β_{SMB}	0.07 [3.09]	-0.09 [-4.61]	-0.11 [-5.94]	-0.17 [-8.99]	0.03 [1.13]	-0.04 [-1.16]
β_{HML}	-0.02 [-0.52]	-0.03 [-1.32]	0.16 [6.11]	0.26 [10.67]	0.02 [0.66]	0.04 [0.83]
β_{RMW}	0.21 [7.22]	0.32 [11.97]	0.19 [7.48]	0.05 [1.88]	-0.07 [-2.04]	-0.28 [-6.15]
β_{CMA}	-0.22 [-5.02]	0.24 [6.14]	0.04 [1.13]	0.08 [2.11]	0.32 [6.52]	0.54 [8.04]
β_{UMD}	-0.06 [-3.68]	0.00 [0.07]	0.02 [1.34]	-0.04 [-3.49]	0.00 [0.17]	0.06 [2.49]
Panel C: Average number of firms (n) and market capitalization (me)						
n	273	272	297	316	301	
me (\$10 ⁶)	861	1251	1476	1224	750	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.38 [3.64]	0.29 [2.84]	0.23 [2.25]	0.28 [2.88]	0.24 [2.43]
Quintile	NYSE	EW	0.50 [6.44]	0.53 [6.70]	0.46 [6.11]	0.42 [5.51]	0.47 [6.37]	0.44 [5.91]
Quintile	Name	VW	0.34 [3.35]	0.35 [3.45]	0.25 [2.59]	0.18 [1.79]	0.24 [2.57]	0.19 [1.94]
Quintile	Cap	VW	0.27 [2.86]	0.31 [3.23]	0.23 [2.47]	0.19 [1.96]	0.27 [3.06]	0.24 [2.62]
Decile	NYSE	VW	0.45 [3.74]	0.49 [4.04]	0.42 [3.47]	0.27 [2.27]	0.43 [3.69]	0.31 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.83]	0.34 [3.25]	0.26 [2.55]	0.23 [2.23]	0.27 [2.77]	0.24 [2.53]
Quintile	NYSE	EW	0.28 [3.36]	0.32 [3.80]	0.26 [3.16]	0.24 [2.96]	0.25 [3.13]	0.24 [3.01]
Quintile	Name	VW	0.29 [2.92]	0.31 [3.07]	0.23 [2.32]	0.18 [1.88]	0.23 [2.46]	0.20 [2.13]
Quintile	Cap	VW	0.23 [2.46]	0.27 [2.83]	0.20 [2.13]	0.17 [1.84]	0.24 [2.77]	0.22 [2.53]
Decile	NYSE	VW	0.40 [3.29]	0.44 [3.63]	0.38 [3.13]	0.29 [2.48]	0.40 [3.41]	0.33 [2.90]

Table 3: Conditional sort on size and IFB

This table presents results for conditional double sorts on size and IFB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFB and short stocks with low IFB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFB Quintiles					IFB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [2.08]	0.66 [2.81]	0.49 [2.17]	0.83 [3.61]	1.02 [3.89]	0.49 [4.32]	0.52 [4.48]	0.47 [4.08]	0.38 [3.27]	0.49 [4.28]	0.42 [3.62]
	(2)	0.68 [2.71]	0.70 [3.06]	0.89 [4.18]	0.82 [3.80]	0.95 [4.12]	0.28 [2.34]	0.34 [2.88]	0.28 [2.41]	0.19 [1.58]	0.28 [2.34]	0.20 [1.67]
	(3)	0.70 [3.04]	0.79 [3.78]	0.83 [4.28]	0.81 [4.04]	0.90 [4.19]	0.20 [1.81]	0.27 [2.46]	0.20 [1.88]	0.20 [1.82]	0.21 [1.94]	0.21 [1.93]
	(4)	0.62 [2.96]	0.83 [4.21]	0.68 [3.62]	0.77 [4.08]	0.80 [4.08]	0.18 [1.92]	0.24 [2.51]	0.17 [1.87]	0.13 [1.39]	0.17 [1.81]	0.14 [1.44]
	(5)	0.39 [2.28]	0.51 [3.19]	0.63 [3.92]	0.48 [2.81]	0.73 [4.46]	0.34 [3.05]	0.37 [3.28]	0.28 [2.50]	0.23 [2.03]	0.32 [3.02]	0.28 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFB Quintiles					IFB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	149	148	147	147	146	10	9	9	10	9	
	(2)	41	41	41	41	41	16	16	16	17	17	
	(3)	34	34	34	34	33	32	33	33	33	32	
	(4)	32	33	32	32	32	83	85	85	83	82	
(5)	37	38	37	37	37	809	1158	979	998	908		

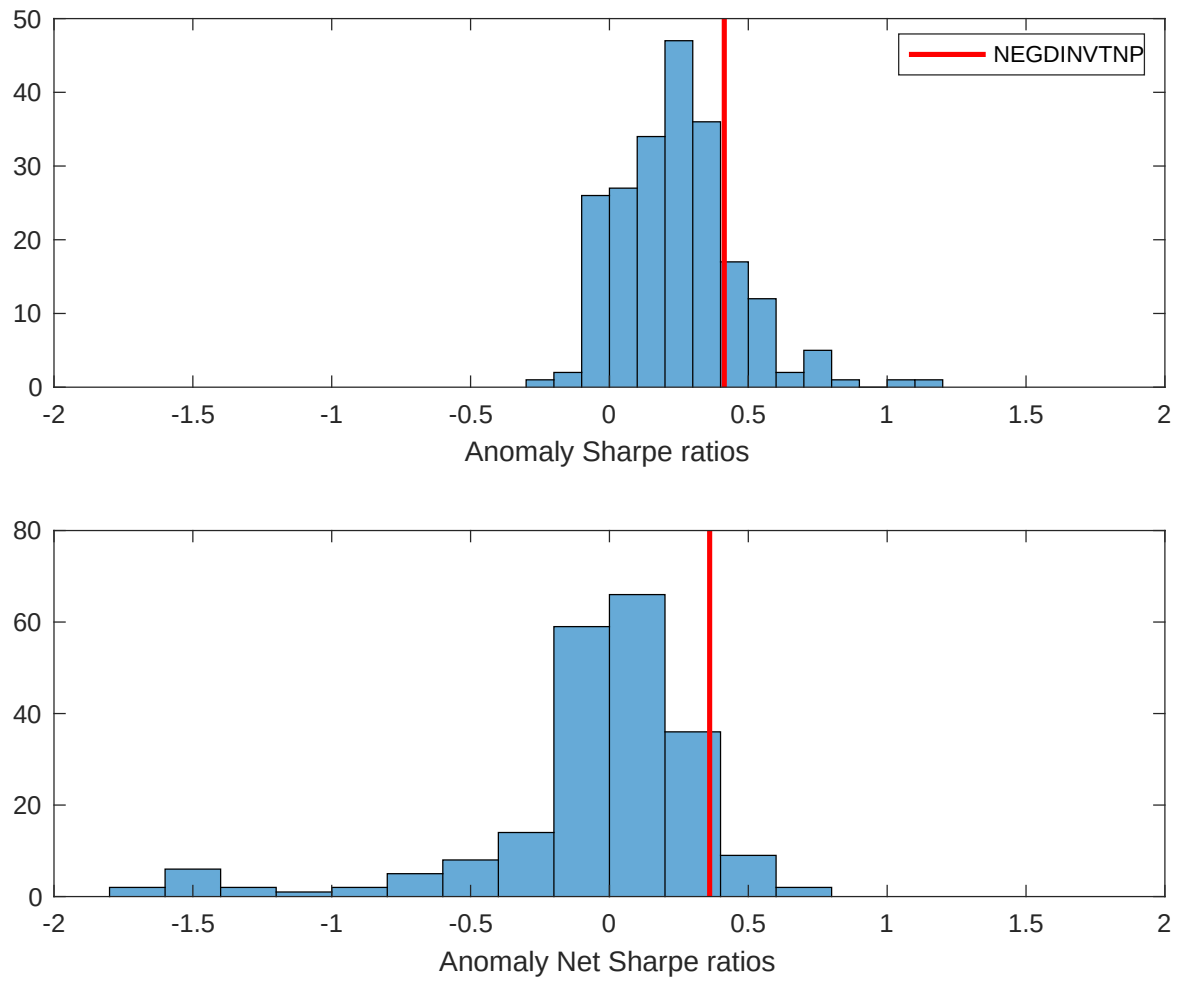


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

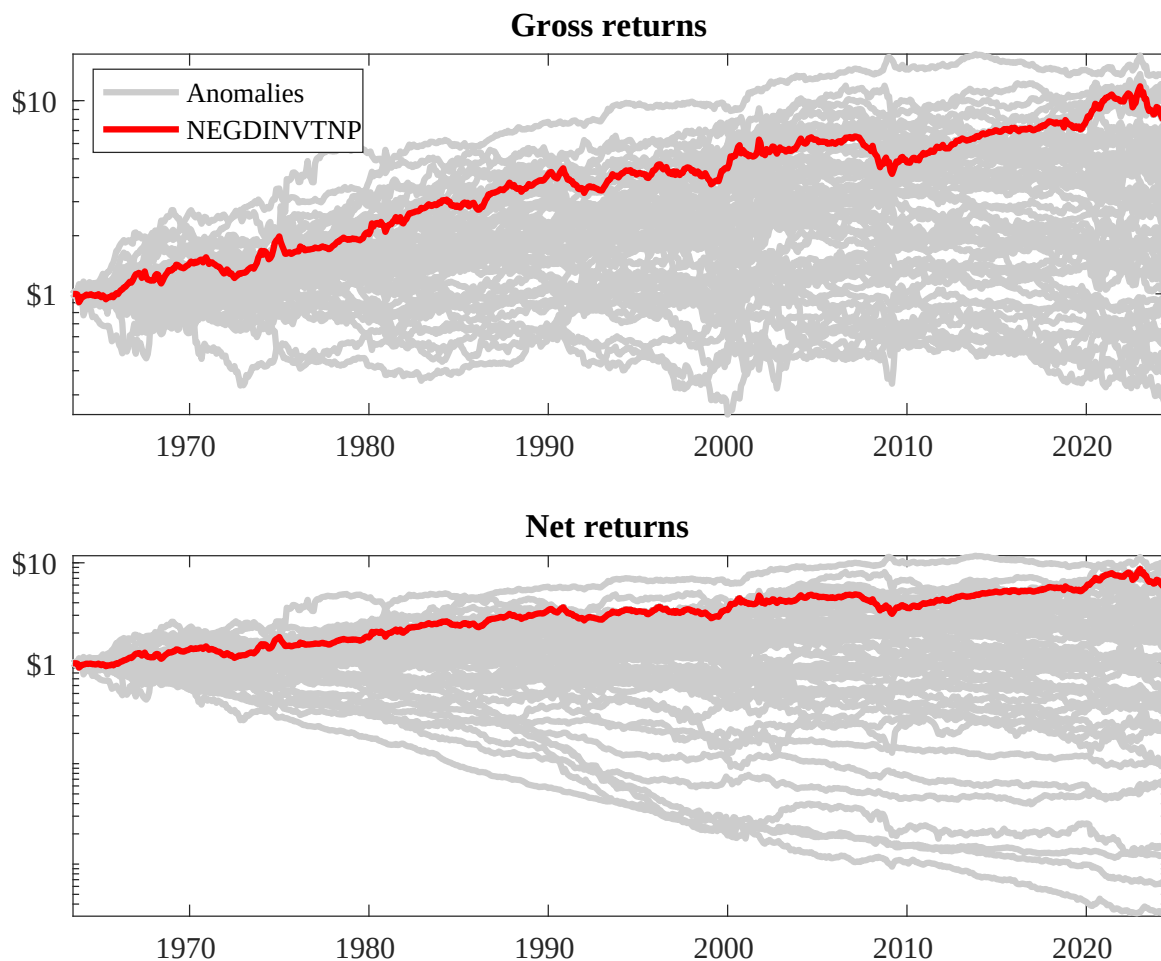
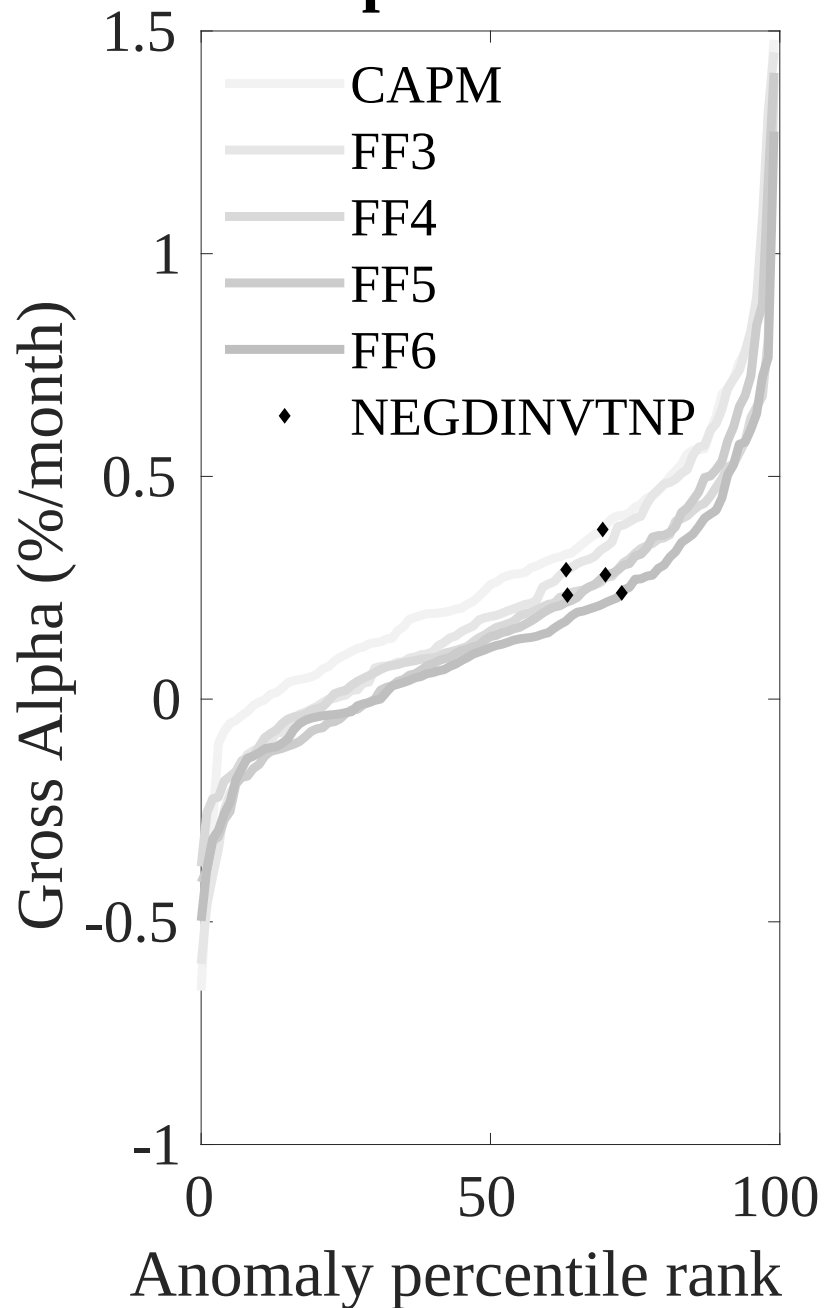


Figure 3: Dollar invested.

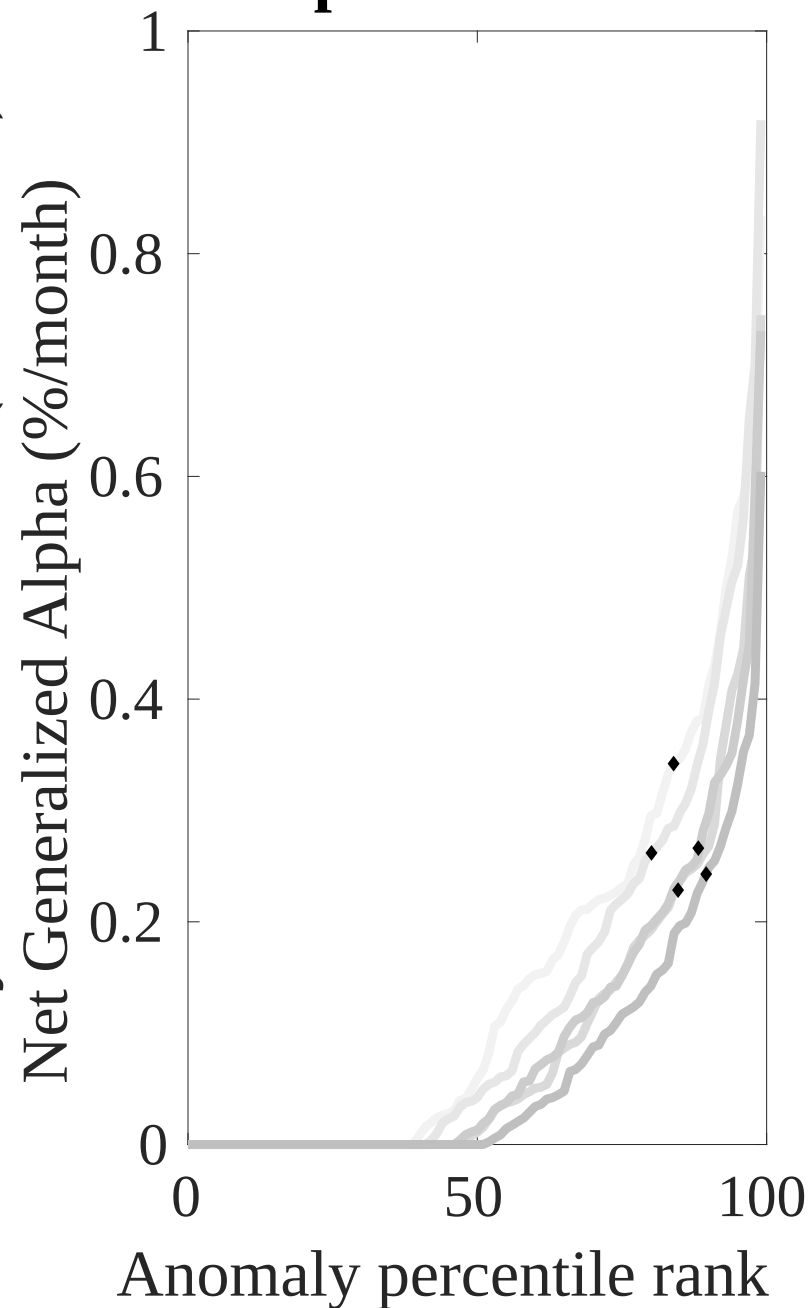
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



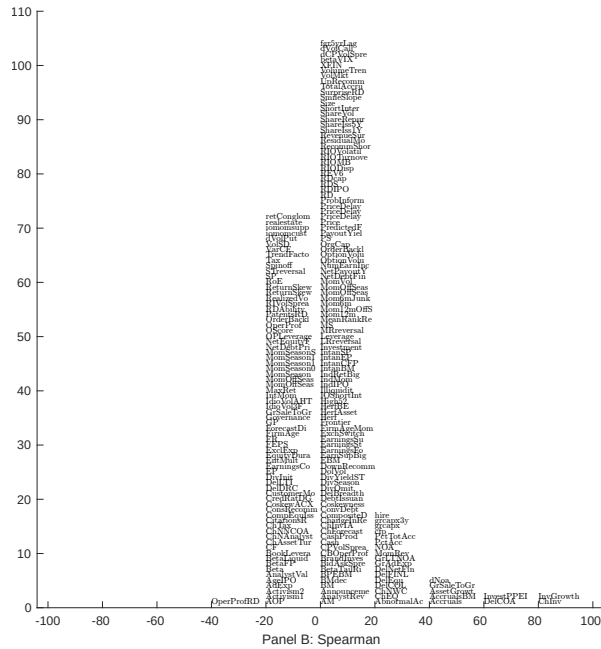
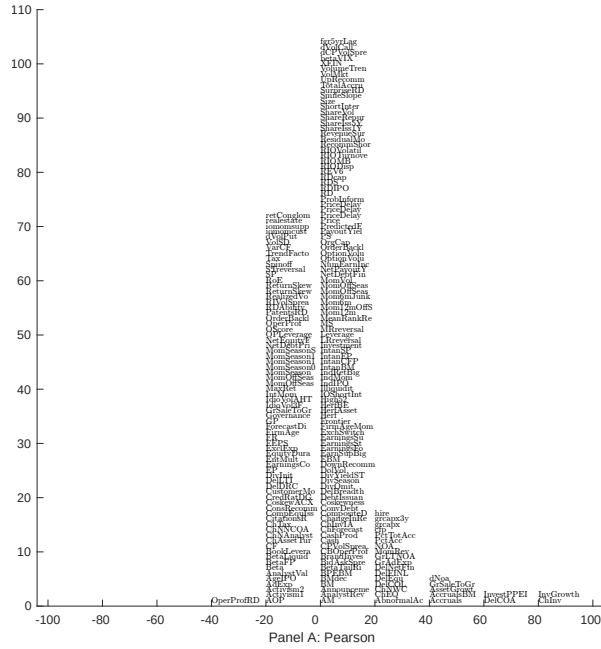


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 203 filtered anomaly signals with IFB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

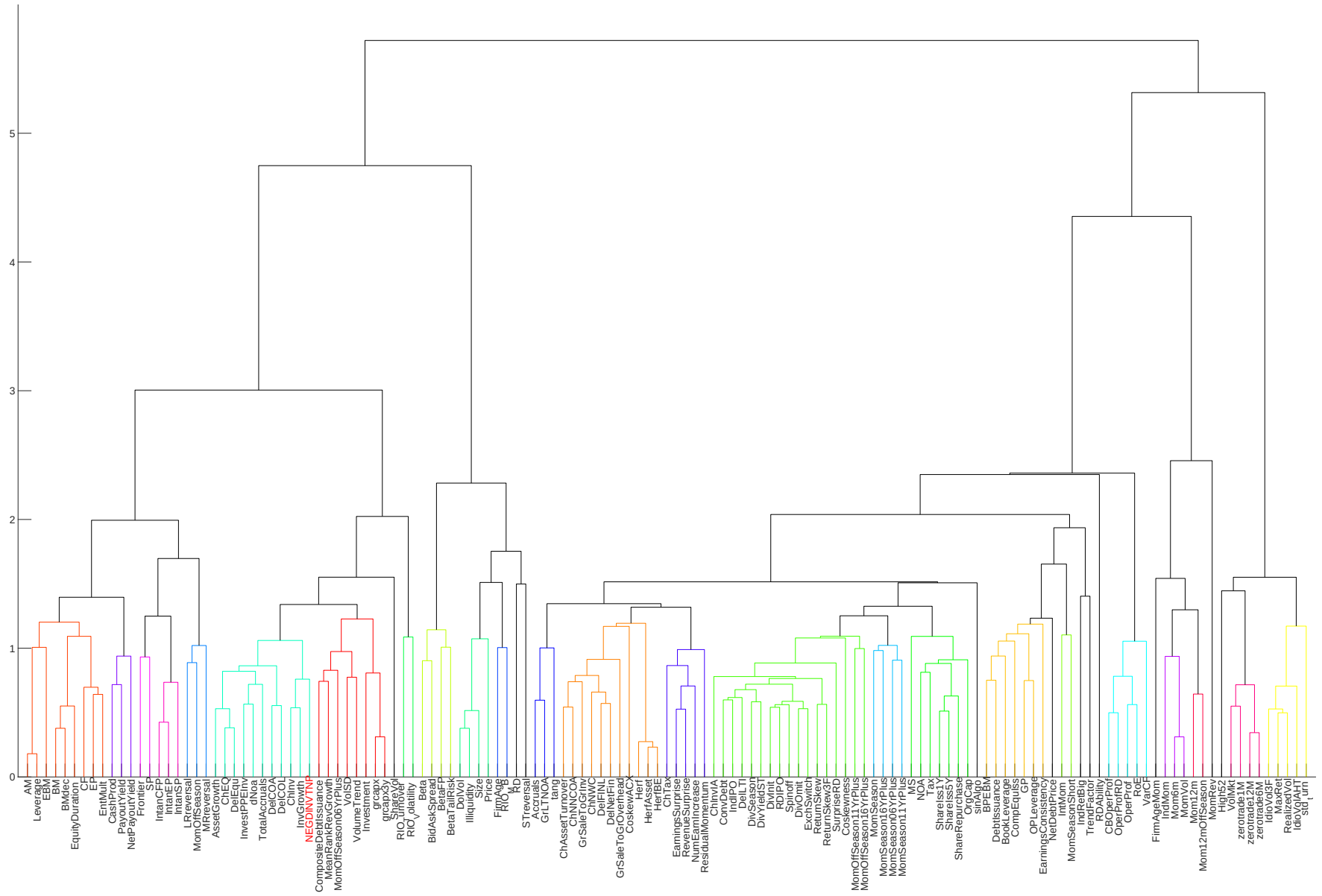


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

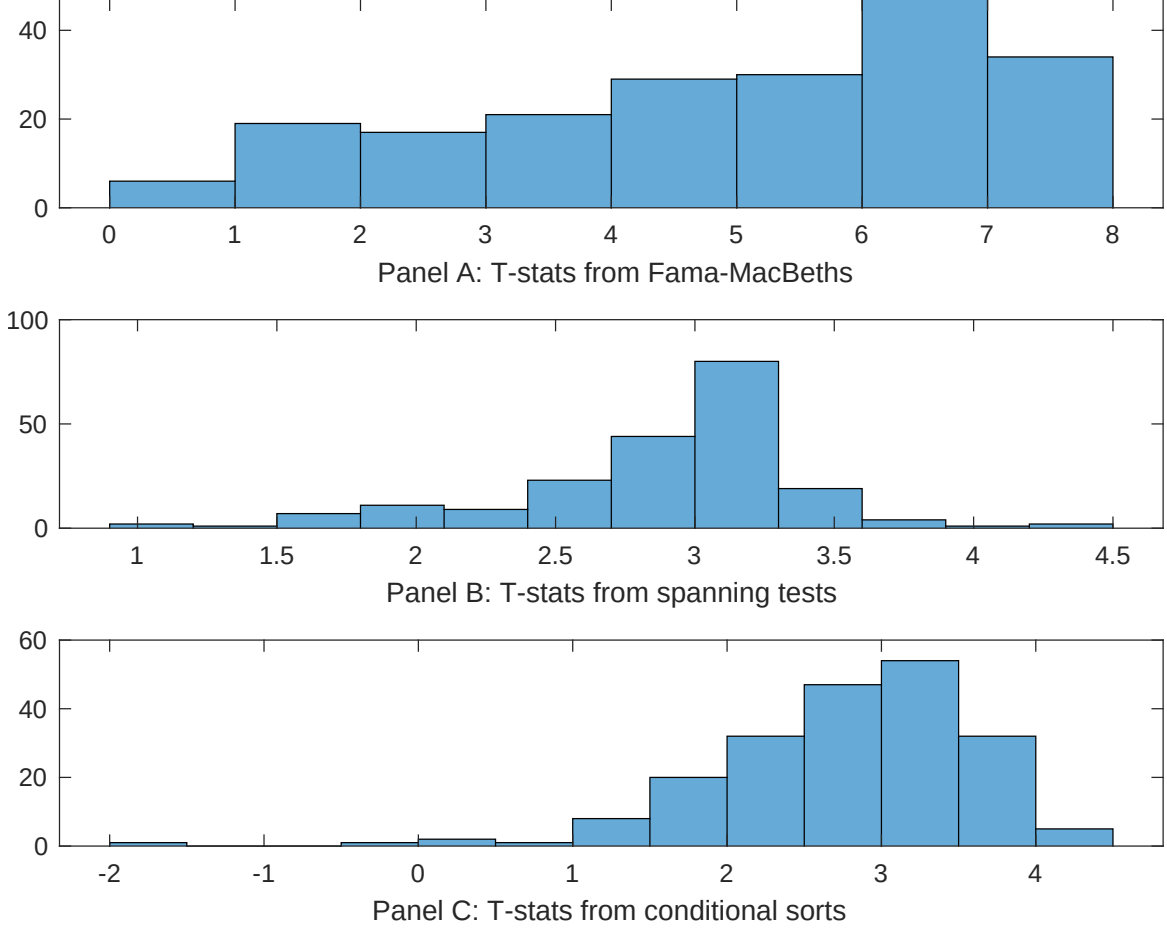


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.79]	0.13 [5.58]	0.13 [6.02]	0.12 [5.81]	0.12 [5.93]	0.12 [5.53]	0.13 [5.55]
IFB	0.68 [3.01]	0.50 [2.17]	0.53 [2.44]	0.83 [3.73]	0.54 [2.42]	0.12 [5.38]	0.28 [1.17]
Anomaly 1	0.39 [6.55]						-0.12 [-1.03]
Anomaly 2		0.76 [8.51]					0.27 [2.08]
Anomaly 3			0.20 [8.58]				0.36 [0.79]
Anomaly 4				0.21 [6.22]			-0.25 [-0.46]
Anomaly 5					0.17 [10.41]		0.13 [4.89]
Anomaly 6						0.11 [6.80]	0.36 [1.55]
# months	732	732	732	732	720	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [1.66]	0.25 [2.78]	0.25 [2.65]	0.26 [2.70]	0.22 [2.24]	0.33 [3.30]	0.21 [2.34]
Anomaly 1	53.82 [12.10]						36.25 [6.36]
Anomaly 2		39.65 [10.73]					18.38 [4.01]
Anomaly 3			30.57 [6.83]				15.47 [3.29]
Anomaly 4				29.55 [6.28]			8.32 [1.68]
Anomaly 5					17.67 [3.21]		-8.39 [-1.46]
Anomaly 6						23.39 [4.35]	-2.70 [-0.50]
mkt	2.96 [1.38]	0.70 [0.32]	0.89 [0.39]	1.07 [0.46]	1.35 [0.58]	1.83 [0.78]	1.89 [0.89]
smb	1.73 [0.55]	-2.13 [-0.67]	-5.98 [-1.81]	1.41 [0.41]	-4.57 [-1.35]	-3.70 [-1.09]	1.60 [0.50]
hml	-0.41 [-0.10]	1.19 [0.28]	-0.16 [-0.04]	-7.93 [-1.66]	1.81 [0.40]	-0.46 [-0.10]	-3.17 [-0.71]
rmw	-19.29 [-4.53]	-23.79 [-5.55]	-28.66 [-6.43]	-25.03 [-5.55]	-28.34 [-6.21]	-27.62 [-6.04]	-19.03 [-4.54]
cma	19.06 [2.82]	21.76 [3.15]	30.84 [4.20]	38.09 [5.44]	41.12 [5.26]	32.81 [4.01]	6.23 [0.78]
umd	2.70 [1.26]	1.68 [0.76]	5.91 [2.61]	7.21 [3.16]	5.60 [2.41]	4.86 [2.08]	2.62 [1.21]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	35	32	26	25	23	23	38

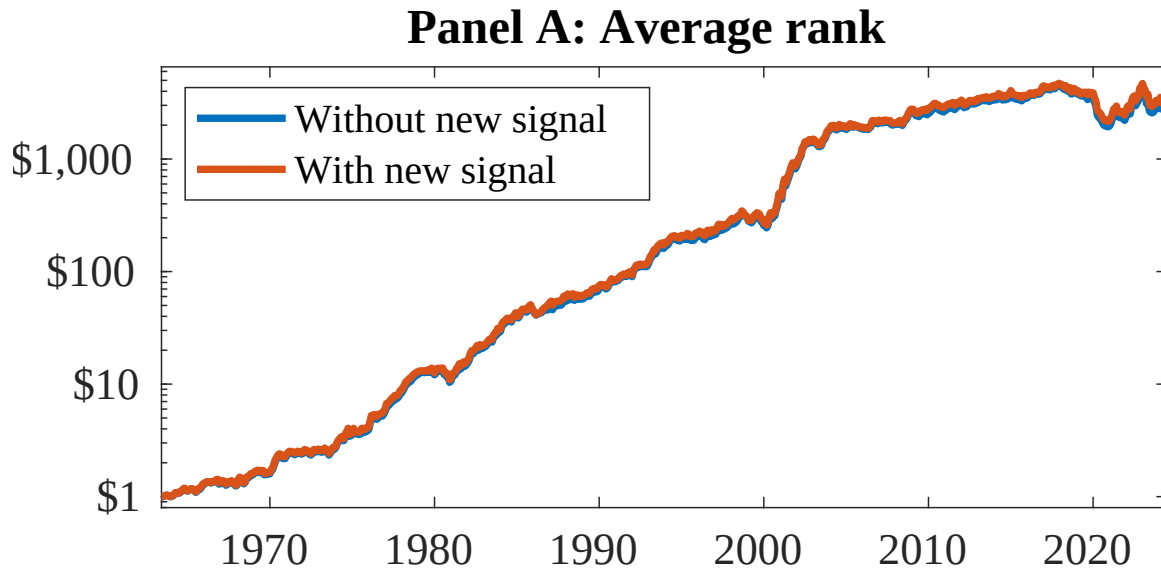


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IFB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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